



پاسخ اولین کاپ برنامه نویسی

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n = 6 # You can change this to test with different values of n
board = [[0 for _ in range(n)] for _ in range(n)]
solutions = 0
col = 0 # Start with the first column
row = 0 # Start with the first row
current_rows = [-1] * n # Keep track of queen positions in each column
while col >= 0:
    # Try to place a queen in the current column
    placed = False
    while row < n:
        # Check if the position is safe (no queens in the same row or diagonals)
        safe = True
        # Check the row and diagonals for all previously placed queens
        for prev_col in range(col):
            prev_row = current_rows[prev_col]
            if prev_row == row or abs(prev_row - row) == abs(prev_col - col):
                safe = False
                break
        if safe:
            board[row][col] = 1 # Place the queen
            current_rows[col] = row # Remember the row where the queen is placed
            placed = True
            row = 0 # Reset row for the next column
            col += 1 # Move to the next column
            break # Exit the row-checking loop
        row += 1
    if not placed:
        # Backtrack if no valid position was found
        col -= 1 # Move back to the previous column
        if col >= 0:
            row = current_rows[col] + 1 # Try the next row in the previous column
            board[current_rows[col]][col] = 0 # Remove the queen
        if col == -1:
            # Found a valid solution
            solutions += 1
            col = -1 # Backtrack to find more solutions
            row = current_rows[col] + 1
            board[current_rows[col]][col] = 0 # Remove the queen
    print(solutions)
```

